Calc III Midterm 2 Review

Spring 2026

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March 26, 2026

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Chapter 1

Definition Review

Definition 1.1 (First order Taylor expansion)**.** Let $f : U \subset \mathbb{R}^n \to \mathbb{R}$ be differentiable at $a \in U$, then

$$f(x) = f(a) + \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(a)(x_i - a_i) + R_1(a, x)$$

where

$$\frac{R_1(a, x)}{\|x - a\|} \to 0 \text{ as } x \to a$$

Definition 1.2 (Second order Taylor expansion)**.** Let $f : U \subset \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable at $a \in U$, then

$$f(x) = f(a) + \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}(a)(x_i - a_i) + \frac{1}{2} \sum_{i,j=1}^{n} \frac{\partial^2 f}{\partial x_i \partial x_j}(a)(x_i - a_i)(x_j - a_j) + R_2(a, x)$$

where

$$\frac{R_2(a, x)}{\|x - a\|^2} \to 0 \text{ as } x \to a$$

Definition 1.3 (critical point)**.** Let $f : U \subset \mathbb{R}^n \to \mathbb{R}$, a point $x_0 \in U$ is a **critical point** of f if either f is not differentiable at x_0, or $Df(x_0) = 0$. A critical point that is not a local extremum is called a saddle point.

Definition 1.4 (quadratic function)**.** A function $g : \mathbb{R}^n \to \mathbb{R}$ is called a **quadratic function** if it is given by

$$g(h_1, \ldots, h_n) = \sum_{i,j=1}^{n} a_{ij} h_i h_j$$

where (a_{ij}) is an $n \times n$ matrix. We can also write g as follows:

$$g(h_1, \ldots, h_n) = [h_1, \ldots, h_n] \begin{bmatrix} a_{11} & \ldots & a_{1n} \\ \vdots & \ddots & \vdots \\ a_{n_1} & \ldots & a_{nn} \end{bmatrix} \begin{bmatrix} h_1 \\ \vdots \\ h_n \end{bmatrix}$$

Definition 1.5 (Hessian matrix)**.** Let $f : U \subset \mathbb{R}^n \to \mathbb{R}$, and suppose all the second-order partial derivatives $\frac{\partial^2 f}{\partial x_i \partial x_j}$ exist, then the Hessian matrix of f is the $n \times n$ matrix given by

$$Hf = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n x_1} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n} \end{bmatrix}$$

The Hessian as a quadratic function is defined by

$$Hf(x)(h) = \frac{1}{2} \begin{bmatrix} h_1 & \dots & h_n \end{bmatrix} Hf(x) \begin{bmatrix} h_1 \\ \dots \\ h_n \end{bmatrix}$$

where $h = (h_1, \dots, h_n)$.

Definition 1.6 (degenerate/nondegenerate points)**.** Let $f : U \subset \mathbb{R}^2 \to \mathbb{R}$ be of C^2, let (x_0, y_0) be a critical point. We define the **discriminant**, $\boldsymbol{\mathcal{D}}$,of the Hessian by

$$\mathcal{D} = \det(Hf) = \left(\frac{\partial^2 f}{\partial x^2}\right)\left(\frac{\partial^2 f}{\partial y^2}\right) - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2$$

If $\mathcal{D} \neq 0$, the critical point (x_0, y_0) is called **nondegenerate**; if $\mathcal{D} = 0$, the point (x_0, y_0) is called **degenerate**.

Definition 1.7 (positive, negative-definite)**.** A quadratic function $g : \mathbb{R}^n \to \mathbb{R}$ is called **positive-definite** if $g(h) \geq 0$ for all $h \in \mathbb{R}^n$ and $g(h) = 0$ implies $h = 0$. Similarly, g is **negative-definite** if $g(h) \leq 0$ for all $h \in \mathbb{R}^n$ and $g(h) = 0$ implies $h = 0$.

Definition 1.8 (global extremum)**.** Let $f : A \to \mathbb{R}$ be a function defined on $A \subset \mathbb{R}^2$ or $A \subset \mathbb{R}^3$. A point $x_0 \in A$ is said to be an **absolute maximum** if $f(x_0) \geq f(x)$ for all $x \in A$. Similarly, x_0 is an **absolute minimum** if $f(x_0) \leq f(x)$ for all $x \in A$.

Definition 1.9 (bounded set)**.** A set $A \subset \mathbb{R}^n$ is said to be **bounded** if there is a number $M > 0$ such that $\|x\| \leq M$ for all $x \in A$.

Definition 1.10 (acceleration)**.** Let $c(t)$ be a path, the **acceleration** $a(t)$ of $c(t)$ is

$$a(t) = c''(t)$$

Definition 1.11 (arc length)**.** Let $c(t) = (x(t), y(t), z(t))$ be a path, then the length of the path in $\mathbb{R}^3$ from $t_0 \leq t \leq t_1$ is

$$\begin{aligned} L_{t_0 \to t_1}(c) &= \int_{t_0}^{t_1} \left(x'(t)^2 + y'(t)^2 + z'(t)^2\right)^{\frac{1}{2}} dt \\ &= \int_{t_0}^{t_1} \|c'(t)\| dt \end{aligned}$$

More generally, if $c(t) = (x_1(t), \ldots, x_n(t))$ is a path in $\mathbb{R}^n$, then

$$L_{t_0 \to t_1}(c) = \int_{t_0}^{t_1} \left(\sum_{i=1}^{n} x_i'(t)^2\right)^{\frac{1}{2}} dt$$

Definition 1.12 (vector field)**.** A vector field is a function $F : A \subset \mathbb{R}^n \to \mathbb{R}^n$ that assigns $x \in \mathbb{R}^n$ to another vector $F(x) \in \mathbb{R}^n$.

Definition 1.13 (flow line)**.** If F is a vector field, a **flow line** for F is a path $c(t)$ such that

$$c'(t) = F(c(t))$$

Intuitively speaking, flow lines are the "streamlines" threading through vector fields.

Definition 1.14 (divergence)**.** Let F be a vector field in $\mathbb{R}^3$ $F = (F_1, F_2, F_3)$, the divergence of F is the **scalar field** (assigns one number to an given point (x, y, z)),

$$\text{div}F := \nabla \cdot F = \frac{\partial F_1}{\partial x} + \frac{\partial F_2}{\partial y} + \frac{\partial F_3}{\partial z}$$

More generally, if $F = (F_1, \ldots, F_n)$ is a vector field on $\mathbb{R}^n$, its divergence is

$$\text{div}F = \sum_{i=1}^{n} \frac{\partial F_i}{\partial x_i} = \frac{\partial F_1}{\partial x_1} + \cdots + \frac{\partial F_n}{\partial x_n}$$

Definition 1.15 (curl)**.** Let F be a vector field in $\mathbb{R}^3$, writing $F = (F_1, F_2, F_3)$, the **curl** of F is the vector field

$$\text{curl}F := \nabla \times F = \det \begin{bmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ F_1 & F_2 & F_3 \end{bmatrix}$$

If $\text{curl}F = 0$, then we say the vector field is **irrotational**.

Definition 1.16. We say a region $D \subset \mathbb{R}^2$ is ***y*-simple** if there are continuous functions ϕ_1, ϕ_2 such that D is the set of pionts (x, y) satisfying

$$x \in [a, b], \quad \phi_1(x) \leq y \leq \phi_2(x)$$

Similarly, we define D to be ***x*-simple** if there are continuous ψ_1, ψ_2 such that D is the set of points (x, y) satisfying

$$y \in [c, d], \quad \psi_1(y) \leq x \leq \psi_2(y)$$

A **simple** region is one that is both x- and y-simple.

Definition 1.17 (injective, surjective)**.** Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be a map, we say T is **injective**, or **one-to-one**, on D^*, if for $x, y \in D^*$

$$Tx = Ty$$

implies

$$x = y.$$

We say T is **surjective**, or **onto** D, if for all $y \in D$, there exists x in the domain of T such that

$$Tx = y$$

If T is both injective and surjective, then we say T is **bijective**.

Definition 1.18 (Jacobian Determinant)**.** Let $T : D^* \subset \mathbb{R}^2 \to \mathbb{R}^2$ be of C^1 defined by

$$T : \begin{pmatrix} u \\ v \end{pmatrix} \mapsto \begin{pmatrix} x(u, v) \\ y(u, v) \end{pmatrix}$$

The **Jacobian determinant** of T, denoted as $\frac{\partial(x,y)}{\partial(u,v)}$ is the determinant of the matrix $DT(u, v)$:

$$\frac{\partial(x, y)}{\partial(u, v)} = \det \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}$$

Chapter 2

Theorem Review

Proposition 2.1 (extremums are critical points)**.** Let $f : U \subset \mathbb{R}^n \to \mathbb{R}$ be differentiable, where U is open. If x_0 is a local extremum, then $Df(x_0) = 0$.

Proposition 2.2 (extremum)**.** Let $f\colon U \subset \mathbb{R}^n \to \mathbb{R}$ be in C^3, and x_0 is a critical point of f. If the Hessian $Hf(x_0)$ is positive-definite, then x_0 is a local minimum of f; if $Hf(x_0)$ is negative-definite, then x_0 is a local maximum.

Proposition 2.3 (local minimum)**.** Let $f(x, y)$ be of C^2, and U is open in $\mathbb{R}^2$. A point (x_0, y_0) is a strict local **minimum** of f if the following conditions hold:

1. $$\frac{\partial f}{\partial x}(x_0, y_0) = \frac{\partial f}{\partial y}(x_0, y_0) = 0$$

2. $$\mathcal{D}(x_0, y_0) > 0$$

   where $\mathcal{D}$ is the **discriminant** of the Hessian, defined by

   $$\mathcal{D} = \det(Hf) = \left(\frac{\partial^2 f}{\partial x^2}\right)\left(\frac{\partial^2 f}{\partial y^2}\right) - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2$$

   where Hf is the 2×2 Hessian matrix.

3. $$\frac{\partial^2 f}{\partial x^2}(x_0, y_0) > 0$$

Proposition 2.4 (local maximum)**.** Let $f(x, y)$ be of C^2, and U is open in $\mathbb{R}^2$. A point (x_0, y_0) is a strict local **maximum** of f if the following conditions hold:

1. $$\frac{\partial f}{\partial x}(x_0, y_0) = \frac{\partial f}{\partial y}(x_0, y_0) = 0$$

2. $$\mathcal{D}(x_0, y_0) > 0$$

   where $\mathcal{D}$ is the discriminant of the Hessian, defined above.

3. $$\frac{\partial^2 f}{\partial x^2}(x_0, y_0) < 0$$

Proposition 2.5 (saddle points)**.** Let $f(x, y) : U \subset \mathbb{R}^2 \to \mathbb{R}$ be of C^2, if $\frac{\partial f}{\partial x}(x_0, y_0) = \frac{\partial f}{\partial y}(x_0, y_0) = 0$, and $\mathcal{D}(x_0, y_0) < 0$, where $\mathcal{D}$ is the discriminant, then the critical point (x_0, y_0) is a saddle point, i.e., neither a maximum or a minimum.

Proposition 2.6 (continuous functions attain extremum on closed bounded sets)**.** Let $f : D \to \mathbb{R}$ be continuous, where D is closed and bounded in $\mathbb{R}^n$. Then f assumes its absolute maximum and absolute minimum values at some point $x_0, x_1 \in D$.

Proposition 2.7. Let f be constrained to a surface S, if f has a max or a min at x_0, then $\nabla f(x_0)$ is perpendicular to S at x_0.

Proposition 2.8 (Lagrange)**.** Suppose that $f : U \subset \mathbb{R}^n \to \mathbb{R}$ and $g : U \subset \mathbb{R}^n \to \mathbb{R}$ are C^1 functions. Let $x_0 \in U$ and $g(x_0) = c$, and let S be the level set for g at c, i.e., $S = \{x : g(x) = c\}$. Assume $\nabla g(x_0) \neq 0$, then if f has a local maximum or minimum on S at x_0, then there exists some real number λ such that

$$\nabla f(x_0) = \lambda \nabla g(x_0)$$

Proposition 2.9 (Bordered Hessian)**.** Let $f : U \subset \mathbb{R}^2 \to \mathbb{R}$ and $g : U \subset \mathbb{R}^2 \to \mathbb{R}$ be smooth functions. Let $x_0 \in U, g(x_0) = c$, and let S be the level curve of g with value c. Assume that $\nabla g(x_0) \neq 0$ and that there exists a real number λ such that

$$\nabla f(x_0) = \lambda \nabla g(x_0)$$

Let $h = f - \lambda g$ and the bordered Hessian determinant is defined by

$$|\bar{H}| = \det \begin{vmatrix} 0 & -\frac{\partial g}{\partial x} & -\frac{\partial g}{\partial y} \\ -\frac{\partial g}{\partial x} & \frac{\partial^2 h}{\partial x^2} & \frac{\partial^2 h}{\partial x \partial y} \\ -\frac{\partial g}{\partial y} & \frac{\partial^2 h}{\partial x \partial y} & \frac{\partial^2 h}{\partial y^2} \end{vmatrix}$$

For f restricted to the curve S,

1. If $|\bar{H}| > 0$, then x_0 is a local max.
2. If $|\bar{H}| < 0$, then x_0 is a local min.
3. If $|\bar{H}| = 0$, then it is inconclusive.

Proposition 2.10 (Newton's Second Law)**.** Let F be the force acting on a particle of mass m, then

$$F = ma$$

where a is the acceleration.

Proposition 2.11 (gradient is irrotational)**.** Let $f \in C^2$, viewing ∇f as a vector field, then

$$\nabla \times (\nabla f) = 0$$

Proposition 2.12 (divergence of a curl vanishes)**.** For any C^2 vector field F,

$$\nabla \cdot (\nabla \times F) = 0$$

Proposition 2.13 (Fubini's Theorem for rectangles)**.** Let f be a continuous function on a rectangular domain $R = [a, b] \times [c, d]$, then

$$\int_a^b \int_c^d f(x, y) dy dx = \int_c^d \int_a^b f(x, y) dx dy$$

Proposition 2.14 (Fubini's Theorem for general regions)**.** Suppose D is a set of points (x, y) such that $y \in [c, d]$ and $\psi_1(y) \leq x \leq \psi_2(y)$, and similarly for $x \in [a, b]$, $\varphi_1(x) \leq y \leq \varphi_2(x)$. If f is continuous on D, then

$$\iint_D f(x, y) dA = \int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} f(x, y) dy dx = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x, y) dx dy$$

Proposition 2.15. We have the following identities regarding divergence and curl:

1. $\nabla(f+g) = \nabla f + \nabla g$.
2. $\nabla(cf) = c\nabla f$, for constant c.
3. $\nabla(fg) = f\nabla g + g\nabla f$.
4. $\nabla(f/g) = (g\nabla f - f\nabla g)/g^2$, at points x where $g(x) \neq 0$.
5. $\text{div}(F+G) = \text{div}F + \text{div}G$.
6. $\text{curl}(F+G) = \text{curl}F + \text{curl}G$.
7. $\text{div}(fF) = f\text{div}F + F \cdot \nabla f$.
8. $\text{div}(F \times G) = G \cdot \text{curl}F - F \cdot \text{curl}G$.
9. $\text{div}\text{curl}F = 0$.
10. $\text{curl}(fF) = f\text{curl}F + \nabla f \times F$.
11. $\text{curl}\nabla f = 0$.
12. $\nabla^2(fg) = f\nabla^2 g + g\nabla^2 f + 2(\nabla f \cdot \nabla g)$.
13. $\text{div}(\nabla f \times \nabla g) = 0$.
14. $\text{div}(f\nabla g - g\nabla f) = f\nabla^2 g - g\nabla^2 f$.

Proposition 2.16 (integrability)**.** For different assumptions on f, we have the following integrability results:

1. Let f be continuous and defined on a closed rectangle R, then f is integrable over R.
2. Let $f : R \to \mathbb{R}$ be a bounded function on R and suppose the set of points where f is discontinuous lies on a finite union of graphs of continuous functions, then f is integrable over R.

Proposition 2.17 (Fubini's Theorem for rectangles). For different assumptions on f, we have the following Fubini's theorem results:

1. Let f be a continuous function on a rectangular domain $R = [a, b] \times [c, d]$, then

$$\int_a^b \int_c^d f(x, y)dydx = \int_c^d \int_a^b f(x, y)dxdy = \iint_R f(x, y)dA$$

2. Let f be bounded with domain $R = [a, b] \times [c, d]$ and the discontinuities of f lie on a finite union of graphs of continuous functions. If the integral $\int_a^b f dy$ exists for each $x \in [a, b]$, then

$$\int_a^b \left(\int_c^d f(x, y)dy \right) dx$$

exists and

$$\int_a^b \int_c^d f(x, y)dydx = \iint_R f(x, y)dA$$

Similar results hold if $\int_a^b f dx$ exists for each $y \in [c, d]$. If both hold simultaneously, then

$$\int_a^b \left(\int_c^d f(x, y)dy \right) dx = \int_c^d \left(\int_a^b f(x, y)dx \right) dy = \iint_R f(x, y)dA$$

Proposition 2.18 (Fubini's Theorem for general regions). Suppose D is a set of points (x, y) such that $y \in [c, d]$ and $\psi_1(y) \leq x \leq \psi_2(y)$, and similarly for $x \in [a, b]$, $\varphi_1(x) \leq y \leq \varphi_2(x)$. If f is continuous on D, then

$$\iint_D f(x, y)dA = \int_a^b \int_{\varphi_1(x)}^{\varphi_2(x)} f(x, y)dydx = \int_c^d \int_{\psi_1(y)}^{\psi_2(y)} f(x, y)dxdy$$

Proposition 2.19 (simple-regions). If D is a x-simple region with $y \in [c, d], \psi_1(y) \leq x \leq \psi_2(y)$, and if f is continuous on D, then

$$\iint_D f(x, y)dA = \int_c^d \left(\int_{\psi_1(y)}^{\psi_2(y)} f(x, y)dx \right) dy$$

Similarly, if D is y-simple, then

$$\iint_D f(x, y)dA = \int_a^b \int_{\phi_1(x)}^{\phi_2(x)} f(x, y)dydx$$

If D is simple, then the two expressions above are equal.

For example, the area of a x-simple region D can be computed as

$$\iint_D dA = \int_c^d \psi_2(y) - \psi_1(y)dy$$

Proposition 2.20. Let A be 2×2 matrix with $\det(A) \neq 0$ and let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the linear map $Tx = Ax$. Then T transforms parallelograms into parallelograms and vertices into vertices.

Proposition 2.21. Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a linear map, i.e., there exists $n \times n$ matrix A such that $Tx = Ax$, then T is injective iff surjective iff $\det(A) \neq 0$.

Theorem 2.1 (change of variables formula). Let D, D^* be elementary regions in $\mathbb{R}^2$, suppose $T : D^* \to D$ is both one-to-one and onto. Then for any integral function $f : D \to \mathbb{R}$, the **change of variable formula** states

$$\iint_D f(x,y)dxdy = \iint_{D^*} f(x(u,v), y(u,v)) \left|\det(J)\right| dudv$$

where

$$\det(J) = \left|\frac{\partial(x,y)}{\partial(u,v)}\right|$$

is the Jacobian determinant.

Proposition 2.22 (change of variables-polar coordinates). As a corollary to the theorem above, we have the following change of variables formula for polar coordinates:

$$\iint_D f(x,y)dxdy = \iint_{D^*} f(r\cos\theta, r\sin\theta)rdrd\theta$$

Proposition 2.23 (change of variables-triple). Let W, W^* be elementary regions in $\mathbb{R}^3$, and suppose $T : W^* \to W$ is bijective. Then the change of varialbes formula for triple integrals states:

$$\iiint_W f(x,y,z)dxdydz = \iiint_{W^*} f(x(u,v,w), y(u,v,w), z(u,v,w)) \mid \det(J) \mid dudvdw$$

where

$$\det(J) = \det \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} & \frac{\partial x}{\partial w} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} & \frac{\partial y}{\partial w} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} & \frac{\partial z}{\partial w} \end{vmatrix}$$

is the Jacobian determinant.

Proposition 2.24 (change of variables-triple cylindrical). As a corollary to the above, we have the following change of variables formula for cylindrical coordinates:

$$\iiint_W f(x,y,z)dxdydz = \iiint_{W^*} f(r\cos\theta, r\sin\theta, z)rdrd\theta dz$$

Recall cylinder cooridnates is setting up the following

$$x = r\cos\theta, y = r\sin\theta, z = z$$

Proposition 2.25 (change of variables-triple spherical). As a corollary to the above, we have the following change of variables formula for spherical coordinates:

$$\iiint_W f(x,y,z)dxdydz = \iiint_{W^*} f(\rho\sin\phi\cos\theta, \rho\sin\phi\sin\theta, \rho\cos\phi)\rho^2\sin\phi d\rho d\theta d\phi$$

Recall the spherical cooridnates is setting up the following

$$x = \rho\sin\phi\cos\theta, y = \rho\sin\phi\sin\theta, z = \rho\cos\phi$$

Chapter 3

Practice Problems